Analytically Integrated Additive Saturating Speckle Models

Disk and Gaussian-Blob Fields over Rectangular Pixels

Abstract

This note defines two synthetic speckle-pattern models designed for analytically integrated image generation: an additive disk model and an additive Gaussian-blob model. Individual blob contributions are integrated over each rectangular pixel, then summed. Intensity scaling, saturation, and bit-depth quantisation are applied only after this integration. The construction is therefore not a Boolean union of disk supports; it is an additive field with a pixel-level saturation nonlinearity. Its purpose is to provide controlled, deterministic image gradients for digital image correlation (DIC) while avoiding numerical pixel quadrature.

1 Pixel-box image formation

Let a rectangular pixel be

$$P = [x_0, x_1] \times [y_0, y_1], \qquad |P| = (x_1 - x_0)(y_1 - y_0). \tag{1}$$

A square pixel of pitch h is the special case $x_1 - x_0 = y_1 - y_0 = h$. For a continuous raw texture field $s(\boldsymbol{x})$, the exact pixel-box average is

$$\bar{s}_P = \frac{1}{|P|} \int_P s(\boldsymbol{x}) \,\mathrm{d}\boldsymbol{x}. \tag{2}$$

The models below use an additive field

$$s(\boldsymbol{x}) = \sum_{i=1}^{N} s_i(\boldsymbol{x}). \tag{3}$$

By linearity,

$$\bar{s}_P = \frac{1}{|P|} \sum_{i=1}^{N} \int_P s_i(\boldsymbol{x}) \,\mathrm{d}\boldsymbol{x}. \tag{4}$$

Thus each blob may be integrated independently, including when blob supports overlap.

1.1 Saturation, intensity mapping, and quantisation

Each blob has peak raw coverage one. After all analytically integrated blob averages have been added, the pixel coverage is saturated:

$$S_P = \mathrm{clamp}(\bar{s}_P, 0, 1)\,, \qquad \mathrm{clamp}(z, 0, 1) = \min(1, \max(0, z))\,. \tag{5}$$

The recorded normalised intensity uses the existing mean and contrast parameters I_0 and γ:

$$I_P = I_0 + \gamma\,(1 - 2S_P)\,. \tag{6}$$

For a B-bit camera, the quantised integer grey level is

$$g_P = \mathrm{round}\big[(2^B - 1) I_P\big]\,. \tag{7}$$

Equations (**??**)–(**??**) are deliberately applied *after* analytic pixel integration. In general,

$$\int_P \mathrm{clamp}(s(\boldsymbol{x}), 0, 1)\,\mathrm{d}\boldsymbol{x} \neq \mathrm{clamp}\left(\int_P s(\boldsymbol{x})\,\mathrm{d}\boldsymbol{x}, 0, |P|\right). \tag{8}$$

The proposed model uses the right-hand ordering as a defined image-generation rule; it is not intended to reproduce the exact geometric union of opaque paint disks.

2 Additive saturating disk model

2.1 Raw disk field

Disk i has centre $\boldsymbol{c}_i = (c_{x,i}, c_{y,i})$, radius R_i, and signed amplitude A_i. Its contribution is

$$s_i^{\mathrm{disk}}(x,y) = A_i\,\mathbf{1}\left[(x-c_{x,i})^2 + (y-c_{y,i})^2 \leq R_i^2\right]. \tag{9}$$

Positive and negative amplitudes permit bright-on-dark or dark-on-bright constructions. The complete field is

$$s^{\mathrm{disk}}(x,y) = \sum_{i=1}^{N} s_i^{\mathrm{disk}}(x,y). \tag{10}$$

Overlap is additive: two coincident unit-amplitude disks produce a raw value of two before the final intensity mapping and saturation.

2.2 Exact rectangular-pixel integral

For disk i,

$$\int_P s_i^{\mathrm{disk}}(x,y)\,\mathrm{d}x\,\mathrm{d}y = A_i\ \mathrm{Area}(P \cap D_i), \tag{11}$$

where D_i is the disk support. Therefore

$$\boxed{\bar{s}_P^{\mathrm{disk}} = \frac{1}{|P|}\sum_{i=1}^{N} A_i\ \mathrm{Area}(P \cap D_i)} \tag{12}$$

is exact, apart from floating-point evaluation of elementary functions.

To evaluate the intersection area, translate the pixel into coordinates centred on disk i:

$$a = x_0 - c_{x,i}, \quad b = x_1 - c_{x,i}, \quad c = y_0 - c_{y,i}, \quad d = y_1 - c_{y,i}. \tag{13}$$

Define

$$h(u) = \sqrt{R_i^2 - u^2}, \qquad |u| \leq R_i. \tag{14}$$

The exact area is the one-dimensional expression

$$\mathrm{Area}(P \cap D_i) = \int_{\max(a,-R_i)}^{\min(b,R_i)} \left[\min(d, h(u)) - \max(c, -h(u))\right]_+ \mathrm{d}u, \tag{15}$$

where $[z]_+ = \max(z, 0)$. Although written as an integral, Equation (**??**) is piecewise analytic. The required circle antiderivative is

$$H(u;R) = \frac{1}{2}\left[u\sqrt{R^2-u^2} + R^2 \arcsin\left(\frac{u}{R}\right)\right], \qquad \frac{\mathrm{d}H}{\mathrm{d}u} = \sqrt{R^2-u^2}. \tag{16}$$

A robust exact implementation proceeds as follows:

1. Clip the horizontal interval to $[a, b] \cap [-R_i, R_i]$.
2. Form breakpoints from the clipped endpoints and every valid intersection of a horizontal pixel edge with the circle:

$$u = \pm\sqrt{R_i^2 - c^2} \quad (|c| < R_i), \qquad u = \pm\sqrt{R_i^2 - d^2} \quad (|d| < R_i). \tag{17}$$

3. Sort and deduplicate the breakpoints.
4. On each open subinterval, evaluate one midpoint to determine which branches of $\min(d, h)$, $\max(c, -h)$, and $[\cdot]_+$ are active.
5. Integrate that fixed branch exactly. Every branch is a linear combination of a constant and $\pm h(u)$, so its definite integral uses only endpoint differences and $H(u; R_i)$ from Equation (**??**).

Useful early-out cases are

$$\begin{aligned} P \cap D_i = \varnothing \quad &\Rightarrow \quad \mathrm{Area}(P \cap D_i) = 0, && (18)\\ P \subseteq D_i \quad &\Rightarrow \quad \mathrm{Area}(P \cap D_i) = |P|, && (19)\\ D_i \subseteq P \quad &\Rightarrow \quad \mathrm{Area}(P \cap D_i) = \pi R_i^2. && (20) \end{aligned}$$

2.3 Why this is not a Boolean union

A Boolean disk pattern would use

$$s_{\mathrm{bool}}(\boldsymbol{x}) = A\,\mathbf{1}\left[\boldsymbol{x} \in \bigcup_i D_i\right], \tag{21}$$

whose pixel integral requires the union area $\mathrm{Area}(P \cap \bigcup_i D_i)$. The additive model instead uses

$$\sum_i A_i \,\mathrm{Area}(P \cap D_i), \tag{22}$$

so overlaps are intentionally counted more than once before saturation. This choice eliminates multi-disk union geometry while retaining exact integration of every individual disk.

3 Additive saturating Gaussian-blob model

3.1 Isotropic Gaussian field

An isotropic Gaussian blob is

$$s_i^{\mathrm{G}}(x, y) = A_i \exp\left[-\frac{(x - c_{x,i})^2 + (y - c_{y,i})^2}{2\sigma_i^2}\right], \tag{23}$$

where A_i is the peak amplitude and σ_i is the standard deviation. The full field is additive:

$$s^{\mathrm{G}}(x, y) = \sum_{i=1}^{N} s_i^{\mathrm{G}}(x, y). \tag{24}$$

Because the exponent is separable in x and y, the rectangular-pixel integral factorises exactly.

Define

$$E_{x,i} = \mathrm{erf}\left(\frac{x_1 - c_{x,i}}{\sqrt{2}\sigma_i}\right) - \mathrm{erf}\left(\frac{x_0 - c_{x,i}}{\sqrt{2}\sigma_i}\right), \tag{25}$$

$$E_{y,i} = \mathrm{erf}\left(\frac{y_1 - c_{y,i}}{\sqrt{2}\sigma_i}\right) - \mathrm{erf}\left(\frac{y_0 - c_{y,i}}{\sqrt{2}\sigma_i}\right). \tag{26}$$

Then

$$\int_P s_i^{\mathrm{G}}(x,y)\,\mathrm{d}x\,\mathrm{d}y = A_i \frac{\pi\sigma_i^2}{2} E_{x,i}E_{y,i}, \tag{27}$$

and hence

$$\boxed{\bar{s}_P^{\mathrm{G}} = \frac{1}{|P|}\sum_{i=1}^{N} A_i \frac{\pi\sigma_i^2}{2} E_{x,i}E_{y,i}} \tag{28}$$

without supersampling or numerical quadrature.

The whole-plane integral of one blob is

$$\int_{\mathbb{R}^2} s_i^{\mathrm{G}}(x,y)\,\mathrm{d}x\,\mathrm{d}y = 2\pi A_i\sigma_i^2. \tag{29}$$

If instead blob i is normalised to prescribed integrated weight w_i, use

$$s_i^{\mathrm{G}}(x,y) = \frac{w_i}{2\pi\sigma_i^2}\exp\left[-\frac{\|\boldsymbol{x}-\boldsymbol{c}_i\|^2}{2\sigma_i^2}\right]. \tag{30}$$

3.2 Axis-aligned anisotropic Gaussian blobs

For widths $\sigma_{x,i}$ and $\sigma_{y,i}$,

$$s_i^{\mathrm{AG}}(x,y) = A_i\exp\left[-\frac{(x-c_{x,i})^2}{2\sigma_{x,i}^2} - \frac{(y-c_{y,i})^2}{2\sigma_{y,i}^2}\right]. \tag{31}$$

The exact pixel integral is

$$\int_P s_i^{\mathrm{AG}}\,\mathrm{d}x\,\mathrm{d}y = A_i\frac{\pi\sigma_{x,i}\sigma_{y,i}}{2}E_{x,i}^{(\sigma_x)}E_{y,i}^{(\sigma_y)}, \tag{32}$$

where each error-function difference uses the corresponding directional width. The whole-plane integral is $2\pi A_i\sigma_{x,i}\sigma_{y,i}$.

A rotated or correlated Gaussian remains analytically integrable over the whole plane, but its axis-aligned rectangular-pixel integral is expressed through a bivariate normal cumulative distribution function rather than a product of two ordinary error functions.

4 Optional compact support for Gaussian blobs

The untruncated Gaussian has infinite support, although its contribution becomes exponentially small. A circular cutoff at $R_i = k\sigma_i$ defines

$$s_{i,k}^{\mathrm{G}}(\boldsymbol{x}) = s_i^{\mathrm{G}}(\boldsymbol{x})\mathbf{1}[\|\boldsymbol{x}-\boldsymbol{c}_i\| \le k\sigma_i]. \tag{33}$$

Its whole-plane integral is

$$2\pi A_i\sigma_i^2\left(1-e^{-k^2/2}\right). \tag{34}$$

However, a circular cutoff destroys Cartesian separability for pixels intersecting the cutoff boundary. Pixels wholly inside the support retain Equation (**??**); pixels wholly outside contribute zero; boundary pixels require a piecewise one-dimensional integral or numerical quadrature. A square cutoff $|x-c_{x,i}| \le k\sigma_i$, $|y-c_{y,i}| \le k\sigma_i$ preserves separability by clipping the pixel limits before evaluating the error-function expression.

Property	Additive disks	Additive Gaussians
Primitive field	Piecewise constant	Smooth
Individual pixel integral	Circle–rectangle area	Product of error functions
Overlap handling	Additive before saturation	Additive before saturation
Compact support	Exact	Optional approximation/cutoff
Boundary regularity	Discontinuous	Infinitely differentiable
Numerical quadrature needed	No	No, without circular cutoff
Main cost	Piecewise geometry	Special-function evaluation

5 Comparison of the two models

Both constructions provide deterministic pixel values and permit direct control of blob centres, sizes, amplitudes, density, mean intensity, contrast, saturation, and bit depth. The Gaussian model offers the simplest closed-form rectangular integral and smooth image gradients. The disk model provides compact, sharp-edged features while remaining analytically integrable through exact disk–rectangle intersection areas.

6 Recommended image-generation sequence

For either model, a clear computational definition is

1. Generate blob parameters $(\boldsymbol{c}_i, R_i, A_i)$ or $(\boldsymbol{c}_i, \sigma_i, A_i)$.
2. Identify blobs whose support or effective support can influence pixel P.
3. Evaluate every individual blob–pixel integral analytically.
4. Sum the contributions and divide by pixel area.
5. Apply affine mean/contrast mapping.
6. Saturate to the camera range.
7. Quantise to the requested bit depth.

This ordering isolates the analytically integrated texture model from the later nonlinear camera operations. It is therefore well suited to renderer-convergence and DIC uncertainty studies in which pixel-integration error must be separated from DIC systematic error.